



Course Video Introduction to Experimental Design & ANOVA Transcript

Hello everyone, welcome to lesson 1 video 1. In this session we will introduce experimental design and analysis of variance. Experimental design provides the framework for collecting data properly and then ANOVA allows us to compare more than two group or group means in a systematic way. We have a lot of cool tools to look at data, but they can't fix one big problem.

You probably have heard saying garbage in garbage out right. Today we will learn how to make sure that we start good. We have all clean data and we do not have garbage.

We'll go over the basics of setting up a good experiment. Then we'll talk about a tool which uses good data and helps us out with different types of results. Okay, so what exactly, why exactly experimental design matter? Imagine you want to know if a fertiliser helps plants grow taller.

A bad way to test this would be to give fertiliser to plants in a sunny spot and no fertiliser to plants in a steady spot right or a shady spot. So you have on one hand plants which are given fertiliser along with sun and on the other hand you also have plants which are not given sun but the fertiliser. So in this simple case what what is likely to happen you might conclude and the first case wherein the plant is in sunlight and also are given fertiliser you might conclude that the plant does are growing up because of the fertiliser right.

So the first case itself is a bad design. The reason for that is you have a sunny spot. Now you cannot differentiate between the effect of sun and the fertiliser.

This entire thing is called as confounding experiment or confounded experiment okay. While a good design would be wherein we isolate the effects, we isolate the sun together in this example okay. So these are the things because of which you might want to do experiments much more cleverly right.

You might want to do experiments such that none of the results are confounded. So this is the language of experiment. You have a response variable which is what you are measuring right.

In this case you are measuring the height of the plant. You have experimental units. These are the kind of things on which we are measuring the entire experiment or doing the entire experiment which in this case are plants.

Now the factor are those which we are testing the plant's heights against right. So these are the explanatory variables or these are the variables which are which we are trying to study the effects for. Now these factors can have different levels.

Levels as in let's say if we are talking about fertilisers, we can have fertiliser A, fertiliser B or no fertiliser C, D and so on right. So levels are nothing but different values of the

factor right. Now these factors are also called as treatments and though let's say if you have three different levels of the factor, in that case you have three treatments.

That's what it means. Now we have three good principles of a good design okay. Now these principles they define what could be a good design okay.

Now to help you understand what do I mean by that is if you think of very simple first term which is called as control because we need something to compare against right. We need some base to compare our results against. So what we do is we create a new level or new group all together which is called as no fertiliser.

So we say that what would happen to plants if they are not given any fertiliser at all right. So that would be our comparisons with those would be our baselines. The very first thing is to establish our baseline okay.

Second is randomisation. We have to assign plants to groups by chance. We cannot assign plants in a defined manner okay.

Like flipping a coin we can flip a coin and decide which plant goes to which fertiliser and so on. This is super important because it helps us mix up all the differences in the plants that we have or it helps us even out various different things okay. Now the very last thing is replication.

This basically is all about keeping or redoing things. Once you have done controlling or set the randomisation you have to repeat the experiment multiple number of times okay. Why this is important though you are doing randomisation there are chances that a lot of things which you do not want in the study might come in and they might destroy your entire experimentation okay.

So that's why replications are important or replicating the entire study again or redoing it again is also very important. Now let's move ahead from taking the data to analysing it right. Look at this chart.

The groups look different. Let's say for example you have control group or you have the baseline here and then you have factor A or fertiliser A and fertiliser B. Now you definitely would be able to make the comparison and say that fertiliser A affects the plant height right. Now imagine that this would have been confounded.

How would you say that whether the increase in height is because of fertiliser A or because of the sunlight right. So all of those things come here and they become important but now we are going to understand how do I differentiate whether the increase in fertilisers the height of plant which were given fertiliser A or when no fertiliser was given whether this increase that you see here right is this significant or not okay and that's the core question is there a statistical significant difference in the average height of the three groups. Now we are moving towards a statistical concept which is used a lot in status x. Okay now the reason I'm saying this is because we have t-test we have z-test those are test included for comparing two groups but

analysis of variance or what we call as ANOVA is used for comparing multiple groups at a time.

Okay you can directly find out whether let's say in this particular example whether the three levels that you have or the three factor levels that you have they are causing actually a difference to the increase in plant's height or not. Okay is there a statistical significant difference between the increase in height. So it will directly tell you whether there is a difference or not.

It will not tell you which particular groups are different but it will tell you whether there is a difference or not which is actually very important and we'll learn about it later on. Okay. So the core idea in ANOVA in analysis of variance is signal versus noise.

Now what is signal? Signal says is there a variability between the groups or let's say are the groups or is the height of plants between the groups different or is there a variability within the group. Okay. Between the group we call that as signal and within the group we call that as noise.

We also call it as the groups or variance between the groups as sum of squares of treatment and variance within the groups as sum of squares of errors. So F-statistics is nothing but the difference between variance between the groups and variance within the groups. Now generally when between the group variance is more right this numerator term is more then F-statistics is likely to be really high and that's what we are looking for.

If F-value is high it means that there is statistical significant difference between the effects of groups or the levels of the factor. Now this is how we set the entire stage. We initially start by stating the hypothesis that all groups are equal.

This is our null hypothesis. Then we say that at least one group is different. Okay.

Then we conclude our F-test which is nothing but the difference between sorry the ratio between signal to noise right variance between the groups divided by variance within the group. Now based on the F-statistics value we deduce p-value if p-value is less than 0.05 then we say that the alternate hypothesis is true otherwise we say that we fail to reject the null hypothesis. So this is what we generally do.

In the next session we will go through what and how to conclude or what how to design the entire experiment to test for ANOVA and we look into ANOVA. But in this let's say in this simple simulation you have been given two different knobs. One knob is the difference between the group means.

If you increase the difference you will see that the F-statistics also increases. If the difference between the group is 0 the F-statistics or let's say it's in this particular case there is there seems to be difference but let's say that there is no difference. Okay.



In that case F-value would be closer to 0. Now similarly let's say there is some difference. Now also if you would want to see is there a noise between groups or within the groups or not. So if we minimise the noise then the denominator gets minimised.

Remember F-statistics is nothing but the variance between the groups to variance within the group. Now in this particular case because you are minimising the denominator hence F-statistics is likely to increase then you see that the P-value. So this is what you will be doing.

And this is what this is how you you generally do the experimentation. So this is how you would set the stage and deduce the experiment and come up with the outputs and then you will play over the hypothesis do the F-test compute the F-test. If F-statistics is significant enough we will go ahead and look for P-value if P-value is less than 0.05 we can conclude that the alternate hypothesis is true and if P-value is not significant then we will say that we failed to reject alternate hypothesis or we failed to reject null hypothesis.

Okay. I hope this intuition was all clear to everyone. In summary experimental design lays the foundation for valid comparisons and ANOVA provides a statistical tool to evaluate differences among groups.

In the next video we will be digging deeper into one way ANOVA and then learn about how it works step by step.

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